

P³SUM: Preserving Author’s Perspective in News Summarization with Diffusion Language Models

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Abstract

In this work, we take a first step towards designing summarization systems that are faithful to the author’s intent, not only the semantic content of the article. Focusing on a case study of *preserving political perspectives in news summarization*, we find that existing approaches alter the political opinions and stances of news articles in more than 50% of summaries, misrepresenting the intent and perspectives of the news authors. We thus propose P³SUM, a diffusion model-based summarization approach controlled by political perspective classifiers. In P³SUM, the political leaning of a generated summary is iteratively evaluated at each decoding step, and any drift from the article’s original stance incurs a loss back-propagated to the embedding layers, steering the political stance of the summary at inference time. Extensive experiments on three news summarization datasets demonstrate that P³SUM outperforms state-of-the-art summarization systems and large language models by up to 13.7% in terms of the success rate of stance preservation, with competitive performance on standard metrics of summarization quality. Our findings present a first analysis of preservation of pragmatic features in summarization, highlight the lacunae in existing summarization models—that even state-of-the-art models often struggle to preserve author’s intents—and develop new summarization systems that are more faithful to author’s perspectives.¹

1 Introduction

What constitutes a faithful summary? In addition to preserving factual consistency—the focus of much prior work (Krzycki et al., 2020; Goyal and Durrett, 2020; Wang et al., 2020a; Pagnoni et al., 2021; Feng et al., 2023a; Tam et al., 2023)—a good summarization system should preserve the *writer’s*

voice—the style, intent, and points of view conveyed by the authors. However, such subtle pragmatic cues are harder to extract and control for by existing models (Borji, 2023), and it remains underexplored whether existing summarization systems generate summaries that are *faithful* to the opinions and perspectives of the authors. Moreover, though language models (LMs) have been widely applied to many summarization tasks, they inevitably contain political biases and such biases could further impact downstream tasks (Feng et al., 2023b). So we hypothesize that summarization systems built on top of LLMs would propagate biases further, but not necessarily align them with stances in the source text. Specifically in the task of summarization, instead of “de-biasing” and generating only neutral summaries, we argue that a good summarization system should *preserve the perspectives* of the authors in generated news summaries.

To this end, we first evaluate to what extent summarization systems and LLMs preserve political stances in generated summaries, by employing a state-of-the-art political perspective evaluator (Liu et al., 2022d) to quantify the gap between stances in news articles and summaries. (§2) We identify that existing summarization systems and LLMs *do* alter opinions and perspectives in the original document, resulting in shifting stances in more than 50% of summaries, with around 25% drifting to the partisan extremes (Figure 1). This highlights a new, underexplored concern with current LLMs as they fail to preserve the intents and perspectives of the authors of news documents during summarization, potentially misinforming the readers.

To address this issue, we propose P³SUM, a summarization model aiming to **P**reserve the **P**olitical **P**erspectives of news articles. (§3) P³SUM employs a non-autoregressive diffusion language model with modular control capabilities to steer the generated summary towards the same perspective of the news article. Specifically, we first fine-tune

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¹Code and data are publicly available at <https://github.com/lyh6560new/P3Sum>.

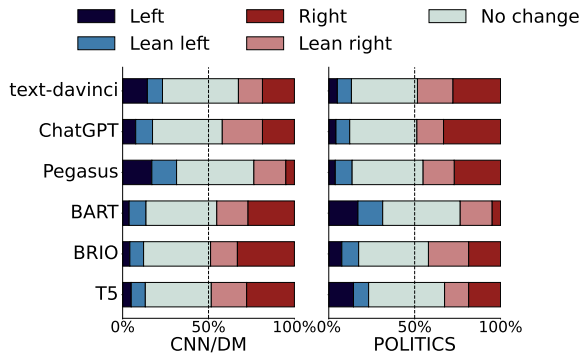


Figure 1: Changes in political stances between the summary and the article. The political perspective classifier produces *left*, *center*, or *right* labels for each text sequence. Left (or Right) indicates a shift in summary stance towards left (or right) by 2 units while Lean Left (Or Lean Right) indicates a shift by 1 unit. No change indicates that there is no difference in the political leaning of the summary and the context. **Our study shows that existing approaches alter the stances of news articles in more than 50% of cases across both datasets.**

a diffusion language model (Mahabadi et al., 2023; Han et al., 2023b,a) on summarization data. During inference, the generated summary is evaluated by a political stance classifier (Liu et al., 2022d) at each step, compared to the target stance in the source document while summary generation is steered towards the target stance. Our primary motivation to use diffusion models is that they allow us to (1) apply the stance classifier on the whole summary at each decoding step, rather than on a prefix generated autoregressively (Kumar et al., 2022b), and (2) seamlessly incorporate various pretrained classifiers without adaptation, to carefully steer generation process. Thus, as an inference-time approach based on diffusion models and controllable text generation (Kumar et al., 2021; Li et al., 2022a; Han et al., 2023a,b; Mahabadi et al., 2023; Austin et al., 2021; Strudel et al., 2022; Dieleman et al., 2022), P³SUM alleviates the need for additional training or pretraining, handles news articles from different ideological stances, and is compatible with future classifiers of author perspectives.

Extensive experiments on three news datasets demonstrate that P³SUM greatly outperforms baselines in preserving the political stances of news articles while maintaining good summarization utility. Specifically, P³SUM is at least 13.7%, 2.9%, and 1.6% better in perspective preservation on CNN/DM (Nallapati et al., 2016), XSUM (Narayan et al., 2018), and POLITICS (Liu et al., 2022d), outperforming popular summarization systems (Raffel et al., 2020; Liu et al., 2022b; Zhang

CHANGE	CNN/DM	XSUM
Left	20.6	5.0
Lean left	13.2	3.8
No change	43.0	39.2
Lean right	15.8	14.2
Right	7.4	37.8

Table 1: Changes (%) in political stances between the gold summary annotations and the news article. Around 57% to 60.8% of reference summaries in news summarization datasets alter author perspectives.

et al., 2020) and large language models (Touvron et al., 2023; Penedo et al., 2023; Chiang et al., 2023). In addition, P³SUM obtains ROUGE scores and abstractiveness metrics that are only slightly lower than state-of-the-art systems, while qualitative analysis highlights P³SUM’s effectiveness in generating high-quality, perspective-preserving summaries. We envision P³SUM as a first step towards summarization systems that are faithful to the intents and perspectives of the authors.

2 Examining Perspective Preservation

Given a news article, the generated summary should preserve the authors’ political perspectives in the document. However, existing models are not designed to control for author intent or perspectives, and we first investigate to which extent summarization systems and large language models alter the perspectives in the generated summaries.

To this end, we measure the political leaning of the generated summaries and compare them to the political stances of original articles, using 500 randomly chosen news articles from the CNN/DM (Nallapati et al., 2016) and POLITICS (Liu et al., 2022d) datasets². We use a political perspective evaluator (Liu et al., 2022d) to quantify political stances of summaries and news articles (mapping text sequences to *left*, *center*, or *right*), investigating the change in political leanings with six summarization models and LLMs: GPT-3.5 (TEXT-DAVINCI-003), CHATGPT (GPT-3.5-TURBO), PEGASUS (Zhang et al., 2020), BART (Lewis et al., 2020), BRIO (Liu et al., 2022b), and T5 (Raffel et al., 2020). We then determine the perspective gap between the summary and the news article.

As shown in Figure 1³, current summarization systems struggle to provide faithful summaries and significantly alter political perspectives. Concretely, the political stance of the generated sum-

²All data are sampled from the test sets of the datasets

³For more specific numbers, please refer to Appendix A

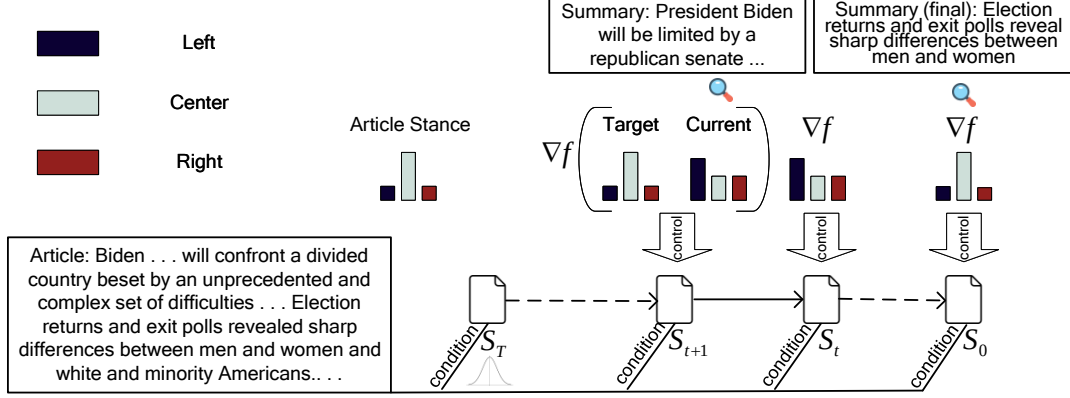


Figure 2: During inference time, we iteratively refine the noisy logits and guide the perspective towards the original political stance by modular control. At each time step, we compare the stance between the current version of the summary and the given article. Then a loss will be calculated if there is any inconsistency, and the corresponding gradients will be backpropagated to steer the generation for the following steps. At training time, we add progressive noise to S_0 and learn to predict S_0 from each noisy S_t .

mary is different from the news article in more than 50% of cases across different models, while around 25% drift to partisan extremes.

Besides, we also examine the political perspective of reference summaries provided in well-established summarization datasets, namely CN-N/DM and XSUM in Table 1, and find that more than 50% of them also alter the stances of the given article. Although these human-written or annotated summaries are considered gold standards for summarization tasks and are used for both training and evaluation, they hardly preserve the original political perspectives, incorporating another layer of data bias into the training and evaluation process.

As a result, how to develop summarization approaches that are faithful to the authors’ perspectives in the news document remains an open research question.

3 P³SUM

We propose P³SUM, a diffusion model that steers the political stance of the generation towards the news article at inference time with an off-the-shelf classifier. Given a news article d , P³SUM aims to generate a summary s that preserves the original political stance of the article. We first finetune a diffusion-based language model on summarization datasets. At decoding time, we employ a political stance classifier to steer the generated summary by incorporating the gradient from the classifier, ensuring that the political stance of the generation is consistent with the original article.

3.1 Diffusion Model Finetuning

At a high level, a diffusion model performs forward diffusion by adding noise to the original data and then learns to reconstruct the input (Sohl-Dickstein et al., 2015; Ho et al., 2020; Chen et al., 2022; Han et al., 2023a,b; Mahabadi et al., 2023). During inference time, we use the learned model to iteratively reconstruct from noisy representations and obtain high-quality generations. To preserve the political stance, we modify the decoding process by incorporating the gradients from an external political classifier iteratively to guide the model generation.

Continuous Data Representation Following Han et al. (2023a), we define a function logits-initialization($\cdot$) to obtain a logits representation over the model’s vocabulary $\mathcal{V}$, mapping each discrete tokens of the news context and summary into continuous space. We map a token w to $\tilde{w} \in \{-K, +K\}^{|\mathcal{V}|}$ as follows:

$$\tilde{w}^{(j)} = \begin{cases} +K & \text{when } w = \mathcal{V}^{(j)} \\ -K & \text{when } w \neq \mathcal{V}^{(j)} \end{cases}$$

where $\mathcal{V}^{(j)}$ denotes the j -th token in the vocabulary and K is a pre-defined scalar hyperparameter.

Forward Diffusion For each passage d and gold summary s , we concatenate them to form a sequence $w = (w_1, \dots, w_L)$. We adopt non-autoregressive modeling (Mahabadi et al., 2023) which feeds the entire sequence into the model to better handle long article contexts. Let $S_0 = (\tilde{w}_1, \dots, \tilde{w}_L) \in \{\pm K\}^{L \times |\mathcal{V}|}$ be the logit representations of w . Each step in the forward diffu-

sion derives $\mathbf{S}_t$ by: $\mathbf{S}_t = \sqrt{\bar{\alpha}_t}\mathbf{S}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon_t$ where $t \in (1, T)$, $\epsilon_t \sim \mathcal{N}(\mathbf{0}, K^2\mathbf{I})$, and $\bar{\alpha}_t \rightarrow 0$ as $t \rightarrow T$ following a predefined schedule. At step T , $\text{sm}(\mathbf{S}_T)$ are fully noisy simplexes over V (we use sm as a shorthand for softmax).

Reverse Process Based on the noisy representation $\mathbf{S}_t$ (or noisy simplex $\text{sm}(\mathbf{S}_t)$) and a current timestep t , we learn to reverse the forward process by predicting the original representation $\mathbf{S}_0$ with our model $\text{Transformer}_\theta$. The predicted outputs are the output logits from the Transformer model θ , denoted as $\hat{\mathbf{S}}_\theta(\mathbf{S}_t, t)$.

$$\hat{\mathbf{S}}_\theta(\mathbf{S}_t, t) = \text{Transformer}_\theta(\text{sm}(\mathbf{S}_t), t) \quad (1)$$

We also apply self-conditioning (Chen et al., 2022) with a 50% probability during prediction, re-computing $\mathbf{S}_t$ in Eq. 1 by:⁴

$$\mathbf{S}_t = \frac{1}{2}(\mathbf{S}_t + \hat{\mathbf{S}}_\theta(\mathbf{S}_t, t))$$

Loss Function After obtaining the model prediction $\hat{\mathbf{S}}_\theta(\mathbf{S}_t, t)$, we employ a cross-entropy loss between this predicted representation of $\mathbf{S}_0$ and the target summary tokens $\mathbf{w}$:

$$\begin{aligned} \mathcal{L}(\theta) &= \mathbb{E}_{t, \mathbf{S}_0} \left[- \sum_{i \in \mathbf{s}} \log p_\theta(w_i | \mathbf{S}_t, t) \right] \\ &= \mathbb{E}_{t, \mathbf{S}_0} \left[- \sum_{i \in \mathbf{s}} \log \text{sm}[\hat{\mathbf{S}}_\theta(\mathbf{S}_t, t)]_{w_i} \right] \end{aligned}$$

where $\log p_\theta(\cdot | \cdot)$ denotes the cross-entropy loss over the output logits of the transformer model θ that we are learning,⁵ and $i \in \mathbf{s}$ denotes whether this token belongs to summary $\mathbf{s}$.

3.2 Perspective-Guided Decoding

A diffusion language model generates the output sequence non-autoregressively by initializing a noise sequence $\mathbf{S}_T$ and iteratively refining it through $\mathbf{S}_{t+1}, \mathbf{S}_t, \dots, \mathbf{S}_0$.

Given an article as input, we initialize the summary as a noisy sequence $\mathbf{S}_T$ where each token is represented as a logit sampled from the normal distribution $\mathcal{N}(\mathbf{0}, K^2\mathbf{I})$. Using our learned model θ , we first obtain an estimated output reconstructing from $\mathbf{S}_T$:

$$\hat{\mathbf{S}}_{\text{sc}, T} = \hat{\mathbf{S}}_\theta(\mathbf{S}_T, T), \quad (2)$$

⁴See Mahabadi et al. (2023) for more details.

⁵For more details, see Han et al. (2023a,b).

Self-Conditioning Mahabadi et al. (2023) observe that self-conditioning (Chen et al., 2022) can improve the consistency between the model predictions and given context. Following their setting, for all steps $t < T$, we perform self-conditioning by mixing and leveraging the predictions from the previous time step in the current step. Let $\mathbf{S}_{t+1}$ denotes the incoming logits at t from the previous time step $t + 1$, and $\hat{\mathbf{S}}_{\text{sc}, t+1}$ denotes the original estimation of the logits at time step $t + 1$. We perform self-conditioning by computing the average of these representations and then pass to the model θ for a prediction:

$$\hat{\mathbf{S}}_{\text{sc}, t} = \hat{\mathbf{S}}_\theta\left(\frac{\mathbf{S}_{t+1} + \hat{\mathbf{S}}_{\text{sc}, t+1}}{2}, t + 1\right)$$

Modular Control We employ political bias classifiers to steer the generated summary toward the stances of the news article. To guide P³SUM to generate summaries with a target political leaning $y \in \{\text{left}, \text{center}, \text{right}\}$, we use an external stance classifier $f_\phi(\cdot)$ that maps texts to the three stance labels and update our previous prediction $\hat{\mathbf{S}}_{\text{sc}, t}$ at each timestep t guided by the gradients from the political stance classifier.

$$\hat{\mathbf{S}}_{\text{ctr}, t} = \hat{\mathbf{S}}_{\text{sc}, t} + \lambda \nabla_{\hat{\mathbf{S}}_{\text{sc}, t}} f_\phi(y | \text{sm}(\hat{\mathbf{S}}_{\text{sc}, t})) \quad (3)$$

where λ is controlling learning rate, a hyperparameter governing the intensity of stance steering and the parameters of ϕ are frozen. This enables P³SUM to iteratively steer the political stances of the generated summary toward the news article. P³SUM employs a modular *plug and control* paradigm so that any off-the-shelf political bias classifier⁶ could be seamlessly integrated.

Logits Projection To obtain the almost one-hot logits similar to the initial data distribution, we further project logits $\hat{\mathbf{S}}_{\text{ctr}, t}$ at the end of every iteration following (Han et al., 2023b):

$$\hat{\mathbf{S}}_{\text{proj}, t}^{(j)} = \begin{cases} +K & \text{if } j = \text{top-}p\text{-sampling}(\hat{\mathbf{S}}_{\text{ctr}, t}) \\ -K & \text{otherwise} \end{cases}$$

where $\text{top-}p$ is the hyperparameter for nucleus sampling (Holtzman et al., 2019). After projecting $\hat{\mathbf{S}}_{\text{ctr}, t}$ to $\hat{\mathbf{S}}_{\text{proj}, t}$, we add a noise according to the forward diffusion schedule and pass the representation $\mathbf{S}_t$ as the incoming logits for the next iteration $t - 1$:

$$\mathbf{S}_t = \sqrt{\bar{\alpha}_t}\hat{\mathbf{S}}_{\text{proj}, t} + \sqrt{1 - \bar{\alpha}_t}\epsilon_t$$

⁶We assume the classifier employs a common tokenizer.

Method	Pres.	Model Size	POLITICS		CNN/DM		XSUM	
			SUC $\uparrow$	DIST $\downarrow$	SUC $\uparrow$	DIST $\downarrow$	SUC $\uparrow$	DIST $\downarrow$
T5	$\times$	200M	44.10	0.35	47.13	0.38	50.53	0.35
BRIO	$\times$	400M	44.95	0.35	48.65	0.37	29.19	0.49
PEGASUS	$\times$	568M	44.19	0.36	44.03	0.37	25.40	0.51
VICUNA	$\times$	7B	52.01	0.30	42.71	0.38	53.19	0.31
FALCON	$\times$	40B	41.51	0.41	40.78	0.39	31.58	0.45
LLAMA2	$\times$	70B	41.97	0.42	43.40	0.39	43.03	0.35
T5	$\checkmark$	200M	47.29	0.34	41.83	0.40	47.97	0.38
BRIO	$\checkmark$	400M	42.15	0.38	46.98	0.38	30.96	0.48
PEGASUS	$\checkmark$	568M	42.38	0.36	43.78	0.38	31.28	0.48
VICUNA	$\checkmark$	7B	53.52	0.29	48.07	0.36	46.02	0.34
FALCON	$\checkmark$	40B	39.64	0.42	46.64	0.36	37.63	0.41
LLAMA2	$\checkmark$	70B	40.15	0.45	43.38	0.44	51.54	0.30
P ³ SUM (ours)	$\checkmark$	125M	54.36	0.28	55.32	0.31	54.75	0.33

Table 2: Performance of political perspective preservation on the three datasets. “Pres.” indicates whether the model is instructed to preserve stances or not. $\uparrow$ and $\downarrow$ indicate whether the metric should be high or low. P³SUM outperforms all baseline models that are 1.6x to 560x larger on five of the six settings across the three datasets.

So the decoding process can be summarized as iteratively denoising logits $\mathbf{S}_T$ to obtain $\mathbf{S}_{t+1}, \mathbf{S}_t, \dots, \mathbf{S}_0$, and $\mathbf{S}_0$ is the final summary. At time step t , we first mix the noisy logits $\mathbf{S}_{t+1}$ and the model estimation $\hat{\mathbf{S}}_{sc,t+1}$ from time step $t+1$ (self-conditioning) and obtain a model estimation for step t : $\hat{\mathbf{S}}_{sc,t}$. Then, we apply the classifier to predict the perspective for the current estimation $\hat{\mathbf{S}}_{sc,t}$ and compare it with a target stance y . The difference between the prediction and the target stance is backpropagated to steer the logits $\hat{\mathbf{S}}_{ctr,t}$. After that, we project the logits $\hat{\mathbf{S}}_{ctr,t}$ to $\hat{\mathbf{S}}_{proj,t}$ and add Gaussian noise to derive $\mathbf{S}_t$. Such process is repeated T times with $\mathbf{S}_0$ as the final representation. The final summary is obtained by converting $\text{argmax } \mathbf{S}_0$ to natural language tokens.

$$\begin{aligned}
\hat{\mathbf{S}}_{sc,t} &= \hat{\mathbf{S}}_{\theta}(\frac{\mathbf{S}_{t+1} + \hat{\mathbf{S}}_{sc,t+1}}{2}, t+1) \\
\hat{\mathbf{S}}_{ctr,t} &= \hat{\mathbf{S}}_{sc,t} + \lambda \nabla_{\hat{\mathbf{S}}_{sc,t}} f_{\phi}(y | \text{sm}(\hat{\mathbf{S}}_{sc,t})) \\
\hat{\mathbf{S}}_{proj,t} &= \text{logits-projection}(\hat{\mathbf{S}}_{ctr,t}) \\
\mathbf{S}_t &= \sqrt{\bar{\alpha}_t} \hat{\mathbf{S}}_{proj,t} + \sqrt{1 - \bar{\alpha}_t} \epsilon_t
\end{aligned}$$

4 Experiments

4.1 Experimental Settings

Datasets We adopt three news datasets: CNN/DM (Nallapati et al., 2016), XSUM (Narayan et al., 2018), and POLITICS (Liu et al., 2022d). Since there are ground truth labels provided in the POLITICS dataset, we directly employ them to measure the performance of preserving perspectives.

Method	POLITICS				CNN/DM			
	R1	R2	R-L	R-avg	R1	R2	R-L	R-avg
T5	38.31	18.04	27.82	33.07	40.82	18.30	28.64	29.25
BRIO	47.91	24.24	33.12	35.09	46.21	22.04	31.36	33.20
PEGASUS	40.62	19.36	29.64	29.87	42.70	19.69	29.76	30.72
VICUNA	21.33	8.84	14.78	14.98	13.20	3.48	8.51	8.40
FALCON	18.77	4.32	11.28	11.46	15.59	3.17	9.43	9.40
LLAMA2	30.93	12.98	20.72	21.54	22.21	6.75	13.89	14.28
P ³ SUM (ours)	37.48	16.50	26.01	26.66	41.12	18.20	27.73	29.02

Table 3: Rouge scores on POLITICS and CNN/DM. Though the decoding process is steered by classifier gradients to preserve political stances, P³SUM’s summarization utility is still competitive among baselines.

Baselines We compare P³SUM with two types of baselines: 1) *summarization systems*, specifically BRIO (Liu et al., 2022b), PEGASUS (Zhang et al., 2020), and T5 (Raffel et al., 2020). 2) *large language models*, specifically Vicuna (Chiang et al., 2023), Falcon (Penedo et al., 2023), and Llama-2 (Touvron et al., 2023).⁷ For each baseline, we employ two modes: *without preservation*, where the baseline is directly used for summarization; *with preservation*, where we prepend instructions to encourage stance preservation.⁸

Implementation We employ the encoder-only ROBERTA-BASE (Liu et al., 2019) as the backbone of P³SUM’s diffusion component. To preserve perspectives at inference time, we leverage the political bias classifier from POLITICS (Liu et al., 2022d), which measures the political stance of the

⁷We test them in the zero-shot setting.

⁸For similar baselines of controllable text generation such as Liu et al. (2021a), we do not compare them with our method since the classifier we use is a discriminator, not a generator as required by the paper.

Context	Model	Summary	Stance
Biden ... will confront a divided country beset by an unprecedented and complex set of difficulties ... Election returns and exit polls revealed sharp differences between men and women and white and minority Americans... His response to these challenges will be limited by a Republican Senate, a solidly conservative Supreme Court majority, hostility from Trump supporters ... Biden enjoyed a big edge with non-white Americans while white voters stuck with the incumbent. ... (center)	Ours	Election returns and exit polls reveal sharp differences between men and women and white...	center ✓
	T5	Biden ... will be limited by a Republican Senate, a solidly conservative Supreme Court majority, hostility from Trump supporters. ...	left ✗
	BRIO	... Biden must confront the pandemic, rebuild the economy and address climate change ...	right ✗

Table 4: A qualitative example of generated summaries from different approaches. Existing summarization systems often alter the political perspective by presenting partial facts or making up non-existing statements. Our method successfully preserves the original perspective by presenting only the main idea and facts in the original article.

generation and compares it with the original stance at each decoding step. This allows a loss term measuring the political stance difference to back-propagate to the embedding layers, penalizing perspective inconsistencies. We provide full details of P³SUM training and inference in Appendix B.

Evaluation We define two metrics to evaluate the success of preserving political stances in the summary using the political stance classifier that maps text sequences to a bias label $f_{bias}(\cdot) : \text{str} \rightarrow \{-1, 0, 1\}$ representing left, center, and right-leaning. 1) *Success Rate* (Suc): $\frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \mathbb{1}(f_{bias}(d) = f_{bias}(s))$, where $\mathbb{1}(\cdot)$ denotes the indicator function and $\mathcal{D}$ denotes the full dataset. 2) *Stance Distance* (Dist): $\frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} |f_{bias}(d) - f_{bias}(s)|$. While Suc examines whether the stance of the summary is consistent with the article, Dist further evaluates how far the perspective of summaries drifts from the news documents. For summarization utility evaluation, we employ Rouge-1/2/L scores (Lin, 2004) and abstractiveness scores (Chan et al., 2021).

4.2 Results

Preserving Author Perspectives Table 2 demonstrates that P³SUM achieves the highest average success rate as well as the lowest stance distance across five of the six settings, outperforming baselines that are 1.6x to 560x larger. For success rate, we surpass the second-best method by 1.6%, 13.7%, and 2.9% respectively on the POLITICS, CNN/DM, and XSUM datasets. This suggests that the combination of diffusion language models and plug-in political bias classifiers offers a promising approach to preserving political perspectives in news summarization.

For large language model baselines that perform

Method	POLITICS	CNN/DM	XSUM
T5	9.02	8.61	7.15
BRIO	5.17	4.11	3.16
PEGASUS	6.76	3.80	6.46
VICUNA	3.98	2.64	1.50
FALCON	1.77	0.83	0.65
LLAMA2	3.99	2.20	1.29
P ³ SUM (ours)	6.32	2.59	2.93

Table 5: Abstractiveness scores (Chan et al., 2021), the lower the better. P³SUM successfully produces concise summaries that are competitive with existing approaches while improving perspective preservation.

Method	CNN/DM
GPT-DAVINCI	0.8444
CHATGPT	0.8935
PEGASUS	0.9395
P ³ SUM (ours)	0.9289

Table 6: Factuality scores (0-1) of the models measured by FactKB (Feng et al., 2023a).

text summarization in a zero-shot setting, we observe that adding instructions for stance preservation produces mixed effects on their performance. For example, the instructions work for FALCON on CNN/DM but are counterproductive on POLITICS. We hypothesize that large language models struggle to grasp the concept of preserving political opinions off-the-shelf, potentially influenced by their internal notion of political leanings that is often biased and inaccurate (Shaikh et al., 2022; Feng et al., 2023b). However, with an explicit classifier-based gradient steering paradigm, P³SUM successfully advances the ability to preserve political perspectives in generated summaries.

Summarization Utility We evaluate P³SUM and baselines on CNN/DM and POLITICS by comparing them to reference summaries⁹ and present

⁹Since no human-written reference summaries are provided in POLITICS, we use LLM-generated summaries se-

results in Tables 3, 6 and 5. Table 3 demonstrates that P³SUM achieves Rouge scores that are on-par with state-of-the-art approaches, while Table 5 shows that P³SUM is producing abstractive and concise summaries. Table 6 suggests that existing approaches and ours are both very factual, echoing the trend of discoveries that summarization systems based on LLMs or their outputs are now overwhelmingly factual, as evaluated by factuality evaluation models¹⁰. Together these results demonstrate that P³SUM gets better at preserving political opinions without greatly sacrificing summarization quality.

Qualitative Analysis In Table 4, we present an example news article from the POLITICS dataset, where models produce summaries with different political leanings. The original article takes a mostly neutral stance, analyzing the electorate and voter issues. However, T5 generates a strongly left-leaning summary by priming the hostility from Republicans and focusing on incorrect facts such as a Republican Senate to support its argument.¹¹ BRIO instead makes a right-leaning pitch by highlighting the challenges looming for the incoming administration. In contrast, P³SUM maintains a neutral standpoint, summarizing the demographic differences in the 2020 election and preserving the original article’s political stance, as confirmed by the stance classifier.

5 Analysis and Discussion

Inherent Bias of Models Previous works suggest that LLMs could have inherent social and political biases (Feng et al., 2023b; Abdulhai et al., 2023; Kurita et al., 2019; Manzini et al., 2019; Cheng et al., 2023; Ladhak et al., 2023). We now explore how LLM inherent biases could prevent models from preserving author perspectives in news summarization. Given center-leaning articles, we take the summaries generated from different systems and measure their political leaning. We then calcu-

lated by humans as reference summaries. To complement the model-generated summaries, we provide an additional human evaluation on the POLITICS dataset in Appendix F.

¹⁰While their factuality scores are high, we find that existing approaches selectively summarize certain sides of the argument, potentially informed by their internal biases. While the information in the summary indeed appeared and is consistent with the news article, it only presents a partial picture of news events and does not account for the full picture. As a result, the focus of our study is stance preservation, aiming to mitigate this partial summarization issue.

¹¹In 2020, Democrats narrowly won control of the senate with a tie-breaking vote from the Vice President.

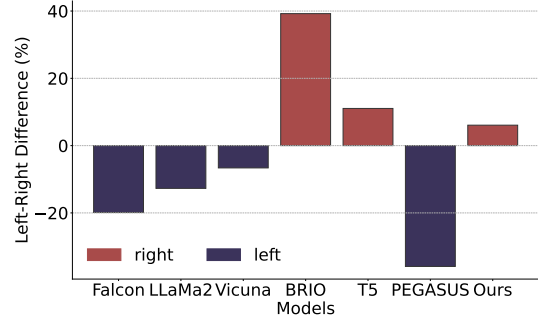


Figure 3: We measure models’ inherent biases by averaging the shift in political stances across all center-leaning articles in POLITICS. P³SUM with explicit controllable generation has the lowest absolute bias.

Ablation	POLITICS		CNN/DM		XSUM	
	SUC↑	DIST↓	SUC↑	DIST↓	SUC↑	DIST↓
P ³ SUM	54.36	0.56	55.32	0.62	54.75	0.65
w/o MC	33.66	0.93	39.53	0.81	52.44	0.69
change	-20.70	+0.37	-15.79	+0.19	-2.31	+0.04
w/o SC	47.36	0.65	44.61	0.78	45.95	0.70
change	-7.00	+0.09	-10.71	+0.16	-8.80	+0.05

Table 7: Ablation study investigating how modular control (MC) and self-conditioning (SC) contribute to P³SUM’s performance.

late the difference between the frequency of right-leaning summaries and left-leaning ones for each model and present the results in Figure 3. Baselines such as BRIO are consistently steering summaries toward the right while most LLMs result in leftward shifts. We argue that these inherent biases present challenges in preserving political perspectives by reinforcing views from one angle, while P³SUM with specific classifier control has the lowest average bias and mitigates these issues.

Effects of Misleading Gold Summary To explore how inconsistent gold summaries can mislead the models, we compare experiments with CHATGPT in the few-shot setting shown in Figure 4. The passage and the corresponding gold summary will be provided first as an example, and then the article will be given again to ask for the model’s summary. We measure how gold summary changes the perspectives of the author and the effects on the model-generated summaries. It is noteworthy that if a reference summary changes the political leaning toward "right" or "lean right", the chance of CHATGPT generating a "right" or "lean right" summary will be improved. And there is a similar trend for the left-leaning examples.

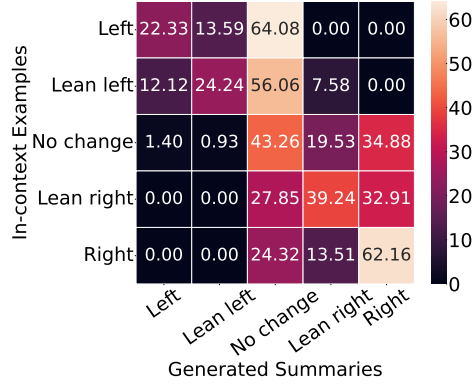


Figure 4: We show how gold summaries as in-context examples alter the perspectives and how model-generated summaries are affected accordingly. We provide CHATGPT with both articles and gold summaries as in-context examples. The left-rightward shift of examples can greatly increase the possibility of similar shifts in the model-generated summaries.

Ablation Study We observe how P³SUM’s performance degrades by dropping the modular control (MC) or self-conditioning (SC) and present the results in Table 7. It is shown that modular control has a significant impact on forcing the model to be faithful to the original opinions. The preserving capacity also drops without self-conditioning.

6 Related Work

Text Summarization and Factuality Evaluation

Research on neural text summarization has produced models and systems that are capable of generating fluent and informative summaries (Liu and Lapata, 2019; Balachandran et al., 2021; Rothe et al., 2021; Narayan et al., 2021; Bhattacharjee et al., 2023; Chen et al., 2023b; He et al., 2023; Liu et al., 2023b; Chen et al., 2023a), given documents from various domains such as news articles (Fabbri et al., 2019; Liu et al., 2022a; Bahrainian et al., 2022), scientific literature (Goldsack et al., 2022), social media and dialogue (Tang et al., 2022; Liu et al., 2022c). However, it remains challenging to generate summaries that are factually consistent with the given document (Cao et al., 2018; Balachandran et al., 2022), resulting in the research area of factuality evaluation. Existing works propose benchmarks to evaluate the factuality of generated summaries (Pagnoni et al., 2021; Tang et al., 2023), develop factuality evaluation models and metrics (Wang et al., 2020b; Kryscinski et al., 2020; Nan et al., 2021; Goyal and Durrett, 2021; Ribeiro et al., 2022; Utama et al., 2022; Laban et al., 2022;

Feng et al., 2023a; Luo et al., 2023), and improve the factuality of generated summaries (Aharoni et al., 2023; Liu et al., 2023a). Recent studies suggest that state-of-the-art large language models (Goyal et al., 2022; Bhaskar et al., 2022) are capable of achieving remarkable factuality in text summarization. However, while LLMs are capable of generating summaries that are factually faithful, our work demonstrates that they struggle to generate summaries that are faithful to the authors’ original opinions and perspectives (Figure 1). As a result, we propose P³SUM, an important first step towards summarization systems that preserve the authors’ opinions in the generated summary.

Understanding the Social and Political Biases of Language Models

Extensive research has demonstrated that machine learning models could encode and exhibit social and political biases (Bender et al., 2021; Jin et al., 2021; Shaikh et al., 2022; Li et al., 2022b). Existing works mainly analyze biases expressed in word embeddings (Bolukbasi et al., 2016; Caliskan et al., 2017; Kurita et al., 2019), token probabilities (Borkan et al., 2019; Bordia and Bowman, 2019; Liu et al., 2021b), model performance discrepancy (Hardt et al., 2016; Feng et al., 2023b), and generated texts (Kumar et al., 2022a). Specifically for political biases, several studies have been proposed to probe LLMs (Bang et al., 2021; Feng et al., 2023b), evaluate the political leaning of texts (Feng et al., 2021; Zhang et al., 2022; Liu et al., 2022d; Qiu et al., 2022), and pretraining LMs on partisan corpora (Jiang et al., 2022). Annotator (Sap et al., 2019, 2022; Gordon et al., 2022) and data bias (Dixon et al., 2018; Dodge et al., 2021; Harris et al., 2022) are commonly attributed as the cause of LM biases, while existing works also established that LM biases could propagate into downstream tasks and cause fairness issues (Li et al., 2020; Feng et al., 2023b; Steed et al., 2022; Ladhak et al., 2023). In this work, we uniquely focus on the task of news summarization: while existing LM-based summarization approaches generate summaries being inconsistent with the political stances of the article, we propose P³SUM to steer the perspective of the summary through iterative controllable generation.

Controllable Text Generation In text summarization, controllable text generation can generate summaries with given entities, predefined lengths, and more (Chan et al., 2021; He et al., 2020; Li et al., 2022a). More generally, inference-time meth-

ods can be used to steer the generation process by altering the output probability distribution at decoding time (Dathathri et al., 2019; Krause et al., 2021; Yang and Klein, 2021; Liu et al., 2021a; Lu et al., 2021; Pascual et al., 2021; Kumar et al., 2021; Qin et al., 2022; Kumar et al., 2022b; Mireshghallah et al., 2022). Particularly, Han et al. (2023a) leverage diffusion-based methods that apply inference-time control through off-the-shelf classifiers. In this work, we further explore the summarization setup using diffusion models to preserve opinions in the decoding process.

7 Conclusion

We demonstrate that existing summarization systems and LLMs struggle to preserve the authors’ political perspectives in news summarization. We present P³SUM, a diffusion-based summarization model that improves political perspective preservation by iteratively guiding the decoding process with an external political stance classifier. Extensive experiments demonstrate that P³SUM outperforms large language models and summarization systems in producing summaries faithful to the political stances of news documents while maintaining competitive summarization utility.

Limitations

Trade off between Utility and Preservation

While P³SUM has achieved state-of-the-art performance in preserving author perspectives among all methods, steering the stance during the inference time can affect the utility of the summary, which results in lower rouge scores or abstractiveness measures. As shown in Figure 1, the gold summaries provided in the datasets do have biases and not the ideal references for preserving original perspectives, which motivates this work and future directions to improve model stability in controllable summarization.

Time Overhead Diffusion models for language are notoriously slower at inference time. While our proposed P³SUM is better than existing summarization systems and LLMs at preserving authors’ political perspectives in the generated summaries, it comes at the cost of inference time subject to the classifier control component at the decoding time of diffusion models. We employ 1000 decoding steps to refine a generated summary so that it is consistent with the news articles’ perspectives and stances, which adds to inference-time compu-

tational costs.

Political Bias Classifier We employ POLITICS (Liu et al., 2022d), an LM-based political bias classifier to iteratively steer the political stances of the generated summary. While it successfully helps to preserve author perspectives, it only provides coarse-grained categorical political leanings (left-/center/right). Besides, it is shown in Liu et al. (2022d) that this political bias classifier is not 100% accurate at identifying political stances, which may mislead the process of preserving the original opinions. Besides, since the classifier we use is based on American political news sources, the political leanings defined in this paper are according to the US policy. There will be different definitions for other countries. However, we argue that our proposed methodology in P³SUM is general and compatible with future political bias classifiers that are more fine-grained, accurate, and appropriate.

Ethics Statement

Although P³SUM’s intended use case is to preserve author perspectives in news summarization, there is a potential risk for misuse of controllable generation models: the same methodology can be used to steer the political leaning of the generated summary towards the hyperpartisan extremes, furthering societal divides and deepening polarization. Therefore, we plan to establish access permission to the fine-tuned P³SUM weights to ensure that it is only used for research purposes.

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CHANGE	TEXT-DAVINCI	CHATGPT	PEGASUS	BART	BRIO	T5
Left	14.4	7.6	17.0	3.8	4.2	5.0
Lean left	8.8	9.8	14.4	9.8	8.0	8.2
Center	44.2	41.2	45.0	40.6	39.0	38.4
Lean right	14.0	18.2	18.6	23.4	15.6	20.6
Right	18.6	27.0	5.0	18.6	33.2	27.8

Table 8: Changes in political stances between the generated summaries and the news article.

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A Changes in Political Stances between Model-Generated Summaries and the Articles

We present specific numbers describing how model-generated summaries shift the author’s points of view in Table 8, as a complement for Figure 1.

B Experiment Details

During fine-tuning on summarization, we use a learning rate of $3e - 5$. We fine-tuned for 20000 steps.

For decoding, we use top- $p = 0.95$ suggested in Han et al. (2023a) and 1000 diffusion steps according to Mahabadi et al. (2023).

Pres.	Prompt
✗	You are an abstractive summarizer that follows the output pattern. Please write a summary for the document. Document:
✓	You are an abstractive summarizer that follows the output pattern. You should always respect the political bias in the original article. Please write a summary for the document. Document:

Table 9: Prompts for large language models, where "Pres." is the abbreviation for whether it belongs to the preservation setting.

Method	POLITICS	CNN/DM
TEXT-DAVINCI	74.93	84.44
CHATGPT	96.15	89.35

Table 10: Factuality scores for LLM-generated summaries.

We implement P³SUM on a server using Tesla V100 GPU with 32 GB memory, 16 CPU cores, and 377GB memory for the experiments.

The backbone of our model is ROBERTA-BASE. It’s noticeable that both P³SUM and the model in (Liu et al., 2022d) use ROBERTA-BASE, and thus they share the same tokenizer. Therefore, as mentioned in (Han et al., 2023a), they can be used for control in an off-the-shelf manner.

For POLITICS, there are no human-written summaries. Therefore, we take the summarization of GPT-TURBO as the ground truth. The details are in the appendix I

With CNN/DM as a popular dataset in text summarization, we aim to test how well P³SUM can perform traditional summarization tasks. However, not all the news articles in the CNN/DM are within the political discipline, which is inappropriate for political leaning preservation. Therefore, we leverage the POLITICS dataset (Liu et al., 2022d), which consists of political news with labels of political leaning.

Hyperparameter	Value
training steps	20000
learning rate	3×10^{-5}
decoding steps	1000
max target length	120
control learning rate λ	4000
simplex value K	5

Table 11: Hyperparameters for P³SUM

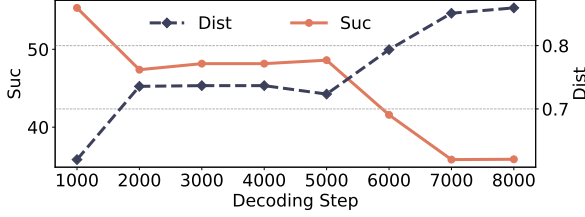


Figure 5: We observe how our model behaves if the total diffusion steps change from 1000 to 8000. If the number of total steps is increased beyond 1000, a drop in the performance would be observed.

C Number of Decoding Steps

Besides control learning rate, another important hyperparameter is the number of decoding steps in the inference time, which can vary from 1000 to 5000 in existing diffusion language models (Han et al. (2023a); Mahabadi et al. (2023)). Thus, we observe how our model behaves if the total diffusion steps change from 1000 to 8000 and present the results in Figure 5. It is shown that the best performance is achieved at step = 1000, and gradually drops when the number of decoding steps increases.

D Stance Control Learning Rate

An important hyperparameter in P³SUM is the classifier control learning rate λ in equation 3, which determines the intensity of stance steering by controlling the gradients. We show how this parameter affects the model’s performance in Figure 6. It is observed that the highest success rate and the lowest distance are achieved at $\lambda = 4000$, and the controlling capability then gradually declines when λ increases, potentially due to top- p setting (Han et al., 2023a).

E Understanding Political Instructions in Prompts

The prompt we use for zero-shot inference for large language models are listed in the Table 9

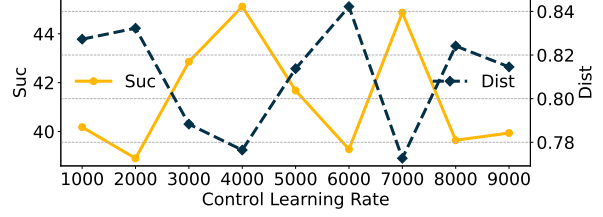


Figure 6: We show how the stance control learning rate λ affects model performance. “Suc” should be high and “Dist” should be low. Best stance preservation is achieved at $\lambda = 4000$, while text degeneration happens with higher λ s.

F Human Evaluation

To complement the manually selected model summaries and the original labels in the POLITICS dataset, we conduct an additional human evaluation to measure the stance preservation and factuality. 150 news articles (50 each for left, center, and right) and their generated summaries are selected from the POLITICS dataset and manually evaluated. The percentage of generated summaries that are faithful to the political stances and facts in the original news article is presented. The results in the Table 13 suggests that human evaluation produces similar results to previous experiments. For example, the average stance preservation rate for our approach is 54.36, while the average with human evaluation is 59.09. This indicates that the current automatic evaluation is a sound solution for scaling such experiments, while we complement it with human evaluation and analysis.

G Ablation Study (cont.)

In addition to success rate and distance, we also present the results of rouge scores for the ablation settings in Table 14.

H Qualitative Analysis (cont.)

Although P³SUM achieves the highest performance on the datasets, it can also fail in certain cases. We present one failure in Table 15 and more examples in the following tables.

I Selecting Criteria

Because there aren’t gold summaries in the POLITICS (Liu et al., 2022d) dataset, we use model-generated summaries for calculating rouge scores. We prompt the TEXT-DAVINCI and CHATGPT, and compare factuality and overall rouge scores.

We calculate the factuality score of summaries by Feng et al. (2023a) and present the scores in

Method	text-davinci as gold				ChatGPT as gold			
	R-1	R-2	R-L	R-avg	R-1	R-2	R-L	R-avg
T5	28.40	11.20	21.66	20.42	36.35	17.50	27.62	27.16
BRIO	31.11	13.66	23.25	22.67	47.91	24.24	33.12	35.09
PEGASUS	26.10	9.40	19.37	18.29	40.62	19.36	29.64	29.87

Table 12: Comparison of rouge scores using TEXT-DAVINCI or CHATGPT as gold summaries.

Category	Left	Center	Right	Avg.
Stance Preservation	63.27	56.00	58.00	59.09
Factuality	48.98	44.00	55.00	49.33

Table 13: Human evaluation results for P³SUM.

Table 10. It is shown that CHATGPT has a higher level of faithfulness.

Choosing TEXT-DAVINCI and CHATGPT as reference summaries respectively, we calculate the rouge scores respectively on POLITICS dataset and

present the results in Table 12.

We can see that most models achieve higher rouge scores when selecting CHATGPT to generate gold summaries, which implies a higher agreement.

Ablation	POLITICS				CNN/DM				XSUM			
	R-1	R-2	R-L	R-avg	R-1	R-2	R-L	R-avg	R-1	R-2	R-L	R-avg
P ³ SUM	37.48	16.50	26.01	26.66	41.12	18.20	27.73	29.02	19.19	2.77	13.08	11.68
w/o MC	36.24	16.21	25.58	26.01	39.66	17.52	27.71	28.29	18.51	2.89	12.35	11.25
change	-1.24	-0.29	-0.43	-0.65	-1.46	-0.69	-0.02	-0.72	-0.68	0.12	-0.73	-0.43
w/o SC	32.60	11.78	21.90	22.09	37.46	13.70	24.89	25.35	19.01	2.53	12.78	11.44
change	-4.88	4.72	-4.11	-4.57	-3.66	-4.50	-2.84	-3.67	-0.18	-0.24	-0.30	-0.24

Table 14: Ablation study (cont.) investigating how modular control (MC) and self-conditioning (SC) contribute to P³SUM’s performance.

Context	Model	Summary	Stance
For months, Republican leaders have been uniform in their insistence that they would allow everyone’s taxes to rise if the rich did not get to keep their Bush-era tax breaks. Mr. Obama has proposed continuing the tax cut for the 98 percent of taxpaying families ... Republicans have demanded tax cuts for all, and, so far, not a single Republican leader has lined up behind Mr. Boehner’s concession. Ultimately, the case for the top-level tax cuts is increasingly shaky. If Republicans are the least bit serious about reducing the deficit, they have to acknowledge that doing so requires additional revenues... (left)	Ours	Republican leaders have been ready to maintain Bush-era tax breaks to continue tax rates. Mr. Obama, who has earned less than \$250,000, will keep up with extra revenue at top rates. ...	right ✗
	T5	The case for the top-level tax cuts is increasingly shaky. If Republicans are the least bit serious about reducing the deficit, they have to acknowledge that doing so requires additional revenues.	left ✓
	BRIO	... Republicans have demanded tax cuts for all, ... If Republicans are serious about reducing the deficit, they have to acknowledge that doing so requires additional revenues.	left ✓

Table 15: Example #1 of one news article, three summaries generated by P³SUM and two baselines, as well as their stances as evaluated by the political bias classifier.

Context	Model	Summary	Stance
Biting his nails nervously, these are the first pictures of the migrant boat captain accused of killing 900 men, women and children in one of the worst maritime disasters since World War Two. Tunisian skipper Mohammed Ali Malek, 27, was arrested when he stepped onto Sicilian soil last night, some 24 hours after his boat capsized in the Mediterranean. Before leaving the Italian coastguard vessel, however, he was forced to watch the bodies of 24 victims of the tragedy being carried off the ship for burial on the island of Malta. He was later charged with multiple manslaughter, causing a shipwreck and aiding illegal immigration. Prosecutors claim he contributed to the disaster by mistakenly ramming the overcrowded fishing boat into a merchant ship that had come to its rescue. ... (left)	Ours	"Mohammed Ali Malek, 27, was arrested when he stepped into Sicilian soil last night.... Mamud Bikhit, 26, also charged with 'aiding illegal immigration' Migrants shifted position on the boat, causing it to overturn as result of collision."	left ✓
	FALCON	The EU’s foreign chief, Federica Mogherini, called for 'rapid, concrete and coordinated' action to deal with the Mediterranean migrant disaster ... She said the situation was a 'humanitarian emergency' and urged EU governments to step up their efforts to help refugees ...	right ✗
	LLAMA2	... The tragedy has sparked widespread outrage and calls for action from politicians and religious leaders. Pope Francis, who has been an outspoken advocate for migrant rights, said he was 'shocked' by the tragedy and called for an end to 'such slaughter'...	left ✓

Table 16: Example #2 of one news article, three summaries generated by P³SUM and two baselines, as well as their stances as evaluated by the political bias classifier.

Context	Model	Summary	Stance
At least 34 people were arrested after hundreds of protesters gathered in New York City on Tuesday night to march against police brutality. The march was organized by National Actions to Stop Murder By Police. Many of the protesters cited the deaths of ... The protesters marched from Manhattan's Union Square and across the Brooklyn Bridge where they partially blocked traffic. Scroll down for video. Hundreds of protesters gathered in New York City on Tuesday night to march against police brutality. ... Organizers say Tuesday's protest was one of 28 being held across the country. Protesters tried to gain access to the roadways of the Brooklyn Bridge, but they were blocked by police. ... Stephen Davis, the Police Department's chief spokesman, said 34 people had been arrested by 6:40 p.m, reports the New York Times. Police say an off-duty police officer driving home on the bridge was assaulted by two protesters when he got out of his vehicle to investigate. Police say the suspects ran off after he identified himself as a police officer. He was hospitalized with injuries to his face and arm. ... (left)	Ours	The march was organized by National Actions to Stop Murder By Police. Protesters cited the deaths of ... Protesters marched from Manhattan's Union Square ... where they partially blocked traffic. Protesters tried to gain access to roadways but were blocked by police. ...	left ✓
	LLAMA2	The protesters marched from Manhattan's Union Square and across the Brooklyn Bridge where they partially blocked traffic. ... Scroll down for video. The protesters marched from Manhattan's Union Square and across the Brooklyn Bridge where they partially blocked traffic. ...	center ✗
	BRIO	The march was organized by National Actions to Stop Murder By Police. ... Many of the protesters cited the deaths of Eric Garner in Staten Island and Walter Scott in South Carolina. Police say an off-duty police officer was assaulted by two protesters on the bridge.	left ✓

Table 17: Example #3 of one news article, three summaries generated by P³SUM and two baselines, as well as their stances as evaluated by the political bias classifier.

Context	Model	Summary	Stance
In Iowa, Ryan says budget a step toward GOP unity. CEDAR RAPIDS, Iowa (AP) — Republican U.S. Rep. Paul Ryan told an Iowa audience Friday that his party can and must come together, and he held out his recently passed budget plan as a sign of growing GOP unity. Although blocs of Republicans object to aspects of the plan passed Thursday in the U.S. House, Ryan said it embodies the principles upon which the nation was founded. "Some people wanted to go further, some people thought it went too far. The point is we unified around these common principles in a plan," the Wisconsin congressman told reporters after headlining a state party dinner in Cedar Rapids. "That's very important to me — which is we can't just oppose, we have to propose." Ryan, the 2012 Republican vice presidential nominee, also played down the significance of his speech in Iowa ... (center)	Ours	U.S. Paul Ryan says his party can and must come together. Ryan says budget plan a step toward GOP unity. Ryan: "Very important to me is we can't just oppose, we have to propose"	center ✓
	FALCON	___ Follow David Pitt on Twitter at ...	✗
	VICUNA	2014 The Associated Press. All rights reserved. This material may not be published, broadcast, rewritten, or redistributed.	✗

Table 18: Example #4 of one news article, three summaries generated by P³SUM and two baselines, as well as their stances as evaluated by the political bias classifier.

Context	Model	Summary	Stance
Chamber of Commerce hits Democratic Senate candidates on union bill. The Chamber has been one of the biggest opponents of EFCA. The group has lobbied heavily against the bill, arguing it would lead to job losses from strikes and union disruption. The labor movement considered the bill one of its top legislative priorities this Congress. Unions say it will help the economy by granting more workers collective bargaining rights, pushing up wages and benefits in the process. Despite the union support, EFCA has struggled to gain enough backing to move past an expected Republican-led filibuster in the Senate. Several Democratic centrists have distanced themselves from the bill, with some, such as Sen. Blanche Lincoln (D-Ark.), coming out against it. So far, EFCA has not had a vote in this Congress. Republicans and business groups, however, remain on alert because Democrats could take up the bill in Congress's lame-duck session after the elections. Union leaders have hinted at that being a possibility. ... (center)	Ours	The Chamber has been one of the biggest opponents of EFCA. The group has argued the bill would lead to job losses from strikes and union disruption. Unions say it will help the economy by granting more workers collective bargaining rights. ...	center ✓
	FALCON	Can you elaborate on the Chamber's argument that EFCA would lead to job losses from strikes and union disruption? How does the labor movement respond to this claim? The Chamber argues that EFCA would allow unions to quickly organize workers ...	left ✗
	LLAMA2	The Chamber of Commerce has been actively campaigning against Democratic Senate candidates due to their support of the Employee Free Choice Act (EFCA). The Chamber believes the bill would lead to job losses and union disruption, while labor unions argue it would help the economy by granting more workers collective bargaining rights and ...	center ✓

Table 19: Example #5 of one news article, three summaries generated by P³SUM and two baselines, as well as their stances as evaluated by the political bias classifier.

Context	Model	Summary	Stance
Nancy Pelosi, Chuck Schumer demand equal airtime for Democrats after Donald Trump address. Democratic leaders on Monday asked the broadcast networks for equal airtime to counter President Trump's planned immigration speech Tuesday, saying it's only fair they get a chance to counter his "malice and misinformation." House Speaker Nancy Pelosi and Senate Minority Leader Charles E. Schumer signaled disappointment with the networks' decisions to grant the president a slot at 9 p.m. ... Democrats must immediately be given equal airtime," they said in a joint statement. The minority party is usually granted time after major address such as a state of the union speech, though reactions to short presidential addresses to the nation are usually less structured. Mr. Trump in his speech is expected to make a plea for Congress to approve ... Democrats have resisted, saying they won't approve any new money for a program they say is unneeded and which Mrs. Pelosi calls "immoral." ... (center)	Ours	Democratic leaders asked the broadcast networks for equal airtime. Nancy Pelosi and Charles E. Schumer signaled disappointed with the networks' decisions to grant the president a slot ... Obama again blamed him for the partial government shutdown ... Democrats have resisted, saying they won't approve any money for a program that is unneeded and which Mrs. Pelosi calls 'immoral'	left ✗
	FALCON	Given that the President is making false claims about border 'security', Mr. Schumer and Mrs. Pelosi said, "we cannot allow the President to use the airwaves, at a time of his choosing, to further mislead the American people." ...	left ✗
	LLAMA2	The president's speech is expected to be carried live on all major television networks, including ABC, CBS, NBC, CNN and Fox News. The speech is scheduled to begin at 9 p.m. EST. ...	✗

Table 20: Example #6 of one news article, three summaries generated by P³SUM and two baselines, as well as their stances as evaluated by the political bias classifier.